📐 SSC CGL — Formulas, Identities & Tricks (ASO Target Post)

> Companion to the Official Syllabus. Scope for **Assistant Section Officer (ASO)** = **Tier-I + Tier-II Paper-I only**. Paper-II (Statistics) and Paper-III (Finance & Economics) are **out of scope** for ASO. Every section maps to the official syllabus para.
>
> **Each table has a "💡 Trick / Shortcut" column.** Diagrams use fixed-width boxes — view in a monospace font (or the generated PDF) for correct alignment.

Legend: `S` =Speed `D` =Distance `T` =Time `P` =Principal `R` =Rate `CP` =Cost Price `SP` =Selling Price · ⭐ = very high-yield.

PART 1 — QUANTITATIVE APTITUDE *(Para 13.10.3 / 13.11.1)*

1.1 Number System & Divisibility ⭐

Concept	Formula	💡 Trick / Shortcut
Sum 1→n	$n(n+1)/2$	last × next ÷ 2
Sum of squares 1→n	$n(n+1)(2n+1)/6$	—
Sum of cubes 1→n	$[n(n+1)/2]^2$	= (sum)²
Sum of first n odd	n^2	"1,4,9,16…"
Sum of first n even	$n(n+1)$	—
No. of factors of $N=a^p \cdot b^q$	$(p+1)(q+1)$	add 1 to each power, multiply
Sum of factors	$(a^{p+1}-1)/(a-1) \times \ldots$	—
HCF × LCM	= product of the two numbers	find one, divide to get other
Divisible by 3 / 9	digit-sum ÷ 3 / 9	add digits
Divisible by 11	(odd-pos sum − even-pos sum) ÷ 11	alternating ±
Divisible by 8	last 3 digits ÷ 8	—
Unit digit of a^n	cycle of 4 (e.g. 2,4,8,6)	take n mod 4

```
Divisibility by 11 :  7  9  2  4   →   (7+2) − (9+4) = 9 − 13 = −4  ✗ not ÷11
                      ↑  ↓  ↑  ↓        (odd places)  (even places)
```

1.2 Percentage ⭐

Concept	Formula	💡 Trick / Shortcut
x% of y	$(x/100)\cdot y$	x% of y = y% of x
% change	$(new-old)/old \times 100$	—
Successive a%, b%	$a + b + ab/100$	both +ve add term; one −ve → subtract
If A is x% more than B	B is $x/(100+x)\times 100$ % less than A	—
If A is x% less than B	B is $x/(100-x)\times 100$ % more than A	—
Value ↑ x% then ↓ x%	net **decrease** = $x^2/100$ %	always a loss
Fraction ⇄ %	1/8=12.5%, 1/6=16.67%, 1/3=33.33%	memorise 1/n table

 Successive sign rule: `a + b + (ab/100)` — plug b as negative for a decrease. Example: 20% up then 20% down → 20 − 20 − 400/100 = **−4%** (net 4% loss).

1.3 Ratio, Proportion & Partnership *(13.11.1.2)*

Concept	Formula	Trick / Shortcut
Proportion a:b=c:d	a·d = b·c	product of means = extremes
Mean proportional of a,b	√(ab)	—
Third proportional of a,b	b^2/a	—
Partnership profit share	∝ (capital × time)	"months-money" product
Componendo-Dividendo	(a+b)/(a−b) = (c+d)/(c−d)	speeds up ratio sums

1.4 Average, Mixture & Alligation ⭐ *(13.11.1.2)*

Concept	Formula	Trick / Shortcut
Average	sum ÷ count	—
New avg on adding 1 item	shifts by (item−avg)/(n+1)	—
Avg speed (equal distance)	$2 \cdot S_1 \cdot S_2/(S_1+S_2)$	**harmonic mean**
Alligation ratio	(Cheaper⇄Mean) : (Dear⇄Mean)	cross-difference
Replacement: remaining	Original·$(1 - x/V)^n$	x removed each of n times, vol V

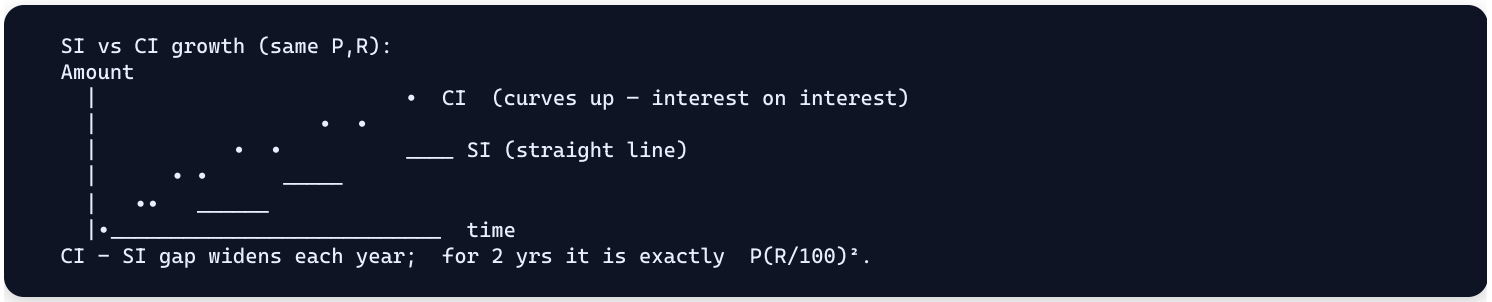


1.5 Profit, Loss & Discount ★ *(13.11.1.2)*

Concept	Formula	□ Trick / Shortcut
Profit %	(SP−CP)/CP × 100	always on **CP**
SP from CP & profit%	CP·(100+P)/100	—
Discount %	(MP−SP)/MP × 100	always on **MP** (marked price)
MP, discount d%, profit p%	CP·(100+p) = MP·(100−d)	links MP↔CP
Two successive discounts	$d_1+d_2 - d_1d_2/100$	same as % successive rule
Dishonest dealer gain%	error/(true−error) × 100	uses false weight
Profit% if SP fixed, CP±x%	use multiplying factors	—

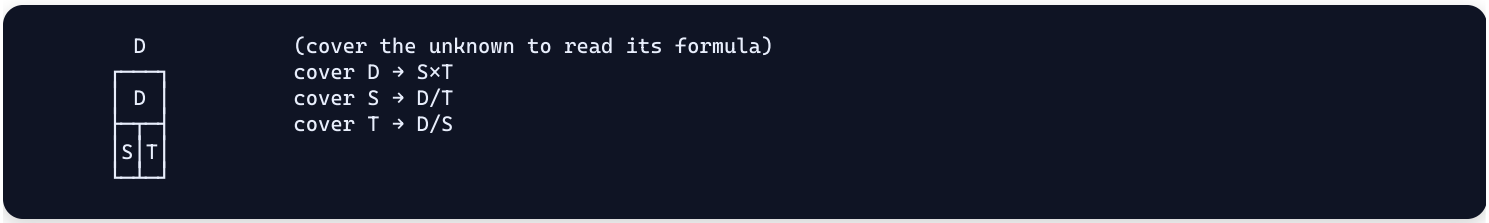
1.6 Simple & Compound Interest ★ *(13.11.1.2)*

Concept	Formula	□ Trick / Shortcut
Simple Interest	P·R·T/100	linear in time
Amount (CI)	$P(1 + R/100)^T$	—
CI − SI (2 years)	$P(R/100)^2$	quick!
CI − SI (3 years)	$P(R/100)^2 \cdot (3 + R/100)$	—
Doubles in (SI)	T = 100/R years	—
Doubles in (CI)	Rule of 72 → 72/R years	approx
Half-yearly CI	rate R/2, time 2T	n compoundings



1.7 Time, Speed, Distance · Trains · Boats ★ *(13.10.3)*

Concept	Formula	□ Trick / Shortcut
Basic	D = S × T	cover triangle below
km/h → m/s	× 5/18	m/s → km/h: × 18/5
Train crossing pole	length ÷ speed	pole = point object
Train crossing platform	(L_train+L_plat) ÷ speed	add both lengths
Two trains opposite	relative S = S_1+S_2	sum
Two trains same dir	relative S = S_1-S_2	difference
Boat downstream	(b + s)	b=boat, s=stream
Boat upstream	(b − s)	—
Boat speed / stream speed	(down+up)/2 / (down−up)/2	half sum / half diff



1.8 Time & Work · Pipes ★ *(13.11.1.2)*

Concept	Formula	□ Trick / Shortcut
Work rate	1/days	A does 1/a per day
A+B together	1/a + 1/b = (a+b)/ab	"product ÷ sum" days
LCM method	Total work = LCM(days); rate = LCM/day	avoids fractions ★
Efficiency ∝	1/time	more efficient = fewer days
$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$	chain rule	men-days-hours-work
Pipe fill − empty	1/fill − 1/empty	leak subtracts

□ **LCM trick:** A=10d, B=15d → Total = LCM 30 units. A=3/day, B=2/day → together 5/day → **6 days**.

1.9 Algebra Identities ★ *(13.11.1.3)*

Identity	Expansion
$(a+b)^2$	$a^2 + 2ab + b^2$
$(a-b)^2$	$a^2 - 2ab + b^2$
$a^2 - b^2$	$(a+b)(a-b)$
$(a+b)^3$	$a^3 + b^3 + 3ab(a+b)$
$(a-b)^3$	$a^3 - b^3 - 3ab(a-b)$
$a^3 + b^3$	$(a+b)(a^2-ab+b^2)$
$a^3 - b^3$	$(a-b)(a^2+ab+b^2)$
$a^3+b^3+c^3-3abc$	$(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
If $a+b+c=0$	then $a^3+b^3+c^3 = 3abc$ ★

Conditional (very high-yield)	Result
$x + 1/x = k$	$x^2 + 1/x^2 = k^2 - 2$
$x + 1/x = k$	$x^3 + 1/x^3 = k^3 - 3k$
$x - 1/x = k$	$x^2 + 1/x^2 = k^2 + 2$

1.10 Geometry — Triangles & Circles ★ *(13.11.1.4)*

📖 **Full handbook with labelled figures for every shape & theorem (lines, triangles, centres, circles, quadrilaterals, polygons, 2D/3D mensuration, coordinate geometry):** Geometry Handbook — Formulas & Figures »

Concept	Formula / Fact	□ Trick
Pythagoras	$a^2 + b^2 = c^2$	triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25
Triangle area	½ · base · height	—
Area (Heron)	√[s(s−a)(s−b)(s−c)], s=perimeter/2	when 3 sides known
Equilateral area	$(\sqrt{3}/4)\cdot a^2$	height = (√3/2)a
Centroid divides median	2 : 1	from vertex
Angle sum	180° (triangle), 360° (quad)	—
Exterior angle	= sum of 2 opposite interiors	—
Inradius r	Area / s	—
Circumradius R	(a·b·c) / (4·Area)	—
Angle in semicircle	90°	diameter subtends right angle
Angle at centre	= 2 × angle at circumference	same arc
Tangent ⊥ radius	90° at point of contact	—

FOUR CENTRES OF A TRIANGLE (don't mix up!)

Centroid (G)	medians meet	ratio 2:1
Incentre (I)	angle bisectors meet	centre of in-○
Circumcentre	perpendicular bisectors	centre of out-○
Orthocentre	altitudes meet	–

Line through G, Circumcentre & Orthocentre = "Euler Line".

1.11 Mensuration 2D / 3D ★ *(13.11.1.5)*

Shape	Area / Volume	Surface / Perimeter
Rectangle	$l \cdot b$	perimeter $2(l+b)$, diagonal $\sqrt{(l^2+b^2)}$
Square	a^2	perimeter $4a$, diagonal $a\sqrt{2}$
Circle	πr^2	circumference $2\pi r$
Triangle	$\frac{1}{2} \cdot b \cdot h$	—
Trapezium	$\frac{1}{2}(a+b) \cdot h$	a,b parallel sides
Cube	$V = a^3$	TSA $6a^2$, diagonal $a\sqrt{3}$
Cuboid	$V = l \cdot b \cdot h$	TSA $2(lb+bh+hl)$, diag $\sqrt{(l^2+b^2+h^2)}$
Cylinder	$V = \pi r^2 h$	CSA $2\pi rh$, TSA $2\pi r(r+h)$
Cone	$V = \frac{1}{3}\pi r^2 h$	CSA πrl, slant $l=\sqrt{(r^2+h^2)}$
Sphere	$V = 4/3 \cdot \pi r^3$	SA $4\pi r^2$
Hemisphere	$V = 2/3 \cdot \pi r^3$	CSA $2\pi r^2$, TSA $3\pi r^2$



1.12 Trigonometry ★ *(13.11.1.6)*

Identity	Value
$\sin^2\theta + \cos^2\theta$	1
$1 + \tan^2\theta$	$\sec^2\theta$
$1 + \cot^2\theta$	$\text{cosec}^2\theta$
$\sin(90-\theta)$	$\cos\theta$ (co-functions)
$\tan\theta$	$\sin\theta/\cos\theta$

```
STANDARD ANGLES TABLE (memorise the sin row, rest follows)
θ        0°     30°     45°     60°     90°
sinθ    √0/2   √1/2    √2/2    √3/2    √4/2
     =   0     1/2    1/√2    √3/2     1
cosθ  = reverse the sin row (1 → 0)
tanθ  = sinθ / cosθ  →  0, 1/√3, 1, √3, ∞

 Trick: write 0,1,2,3,4 under the angles → take √ → divide by 2 = sinθ.
```

Heights & Distances	Quick fact
Angle of elevation 45°	height = distance (tan45=1)
30°	height = distance/√3
60°	height = distance·√3

PART 2 — REASONING & GENERAL INTELLIGENCE *(Para 13.10.1 / 13.11.2)*

2.1 Coding–Decoding

Type	Method	Trick
Letter↔Number	A=1 … Z=26	"EJOTY": E=5,J=10,O=15,T=20,Y=25
Reverse position	A=26 … Z=1	sum with normal = 27
Shift cipher	+n or −n on each letter	watch wrap-around Z→A
Opposite letter	letter + its pair sums to 27	A↔Z, B↔Y (mirror)

```
EJOTY anchor (count from nearest):  A B C D E  ...  J  ...  O  ...  T  ...  Y Z
                                    1 2 3 4 5      10     15     20     25 26
```

2.2 Number / Letter Series

Pattern	Clue
Constant difference	arithmetic (e.g. +5)
Multiply/divide	geometric (×2, ÷3)
Squares / cubes	1,4,9,16 or 1,8,27,64
Prime sequence	2,3,5,7,11,13
Alternating	two interleaved series
±increasing diff	+2,+4,+6 … (2nd difference constant)

2.3 Blood Relations



□ Trick	How
Use symbols	+ male, − female, → married, ⊥ child
Maternal vs Paternal	maternal = mother's side
Self-reference phrases	collapse "the X of Y" to one known person — see below

🔑 Cracking "only son/daughter of his/her ___" (self-reference trick)

The exam hides the answer behind a chain like *"the only daughter of his mother."* **Rule:** replace the phrase from the **inside out**, and use the **pronoun** (his/her) to lock who is speaking.

```
GOLDEN SUBSTITUTIONS (memorise these collapses)
┌──────────────────────────────────────────────┬──────────────────────────────┐
│ Phrase (pointing to a picture/person)        │ Collapses to                 │
├──────────────────────────────────────────────┼──────────────────────────────┤
│ "only son of HIS mother"                     │ = the man HIMSELF            │
│ "only daughter of HIS mother"                │ = his SISTER                 │
│ "only daughter of HER mother"                │ = the woman HERSELF          │
│ "only son of HER father"                     │ = her BROTHER                │
│ "his father's only son"                      │ = he HIMSELF (no brother)    │
│ "my mother's only son's wife"                │ = MY wife (I am that son)    │
└──────────────────────────────────────────────┴──────────────────────────────┘
Why: "only" means no siblings of that type → the description can point to just ONE person.
```

Example 1 — classic photo puzzle

> *Pointing to a photograph, a man said, "She is the **only daughter of my mother**." Who is in the photo?*

- "only daughter of my mother" + pronoun **my** (the man speaks) → his mother's only daughter = **his sister**.

✅ **Answer: his SISTER.**

Example 2 — the two-step chain

> *A woman said, "He is the **only son of the mother of my brother**." How is he related to her?*

- "mother of my brother" = **my mother** (same mother).
- "only son of my mother" = **my brother** (if speaker is female) — but he's already called her brother → confirms it.

✅ **Answer: her BROTHER.**

Example 3 — wife hidden inside

> *Pointing to a lady, Raman said, "Her mother is the **only daughter of my mother**." How is the lady related to Raman?*

- "only daughter of my mother" = Raman's **sister**.
- So the lady's mother is Raman's sister → the lady is the **daughter of Raman's sister** = his **niece**.

✅ **Answer: his NIECE.**

Example 4 — "father's only son" = himself

> *"That boy is the son of my **father's only son**." (spoken by a man) How is the boy related to him?*

- "my father's only son" = the speaker **himself**.
- son of himself = **his son**.

✅ **Answer: his SON.**

```
STEP METHOD (use on every self-reference question)
1) Find the innermost phrase ("X of my/his/her Y").
2) Apply a GOLDEN substitution → replace with ONE person.
3) Re-read the sentence with that person plugged in.
4) Repeat outward until only "speaker → target" remains.
5) Decide gender from his/her/she/he, then name the relation.
```

 Quickest tells: "*only daughter of HIS mother*" is almost always **sister**; "*only son of HER father*" is almost always **brother**; whenever the **pronoun matches the gender of the person described** (his…son, her…daughter), it usually collapses to the **speaker himself/herself**.

2.4 Clocks & Calendars ⭐

📖 **Full topic with derivations, 12 worked examples, diagrams & a practice set:** Clocks & Calendars — Deep Dive »

Concept	Formula	Trick		
Angle between hands	\	30H − 5.5M\	degrees	H=hour, M=minutes
Hands overlap	every 65·5/11 min (~65.45)	11 times in 12 h		
Minute hand gains	5.5° per minute	—		
Odd days in 100 yrs	5	200→3, 300→1, 400→0		
Leap year	÷4 (century ÷400)	2000 leap, 1900 not		
Day code (week)	count odd days from a known date	—		

```
CLOCK ANGLE :  θ = |30·H − 5.5·M|
e.g. 3:40  →  |30·3 − 5.5·40| = |90 − 220| = 130°
(if >180°, subtract from 360° for the smaller angle)
```

2.5 Direction Sense & Dice




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DICE (standard):
opposite faces sum to 7
1⇔6, 2⇔5, 3⇔4
If two dice show same 4 faces,
the two odd faces are opposite.
```

```
Right turn = clockwise (N→E→S→W). Shadow at morning falls West, evening East.
```

Topic	□ Trick
Net displacement	use Pythagoras on net N-S & E-W legs
Syllogism	draw Venn circles; "possibility" cases for "some/all"
Venn diagram	smallest region logic; least/most overlap

PART 3 — COMPUTER KNOWLEDGE / PROFICIENCY (Paper-I, Section-III) ✅ ASO *(Para 13.11.5)*

> In scope for ASO (it sits inside the compulsory Paper-I). Mostly factual — these tables are the high-yield recall points.

3.1 Computer Basics & Memory

Concept	Key fact	□ Trick
CPU parts	ALU + Control Unit + Registers	"brain" of computer
ALU	does arithmetic & logic	A = Arithmetic, L = Logic
Memory units (small→big)	bit < Byte < KB < MB < GB < TB	1 Byte = 8 bits; ×1024 each step
RAM	volatile, temporary, fast	lost on power-off
ROM	non-volatile, permanent	BIOS lives here
Cache	fastest memory, near CPU	speeds up repeat access
Primary vs Secondary	RAM/ROM vs HDD/SSD/pen-drive	secondary = storage
Input devices	keyboard, mouse, scanner, mic	data IN
Output devices	monitor, printer, speaker	data OUT

```
DATA-SIZE LADDER (each ×1024)
bit → Byte → KB → MB → GB → TB → PB
 1     8b   1024B  ...
□ "Big Kids May Get Trophies" → Byte, KB, MB, GB, TB.
```

3.2 Software, MS Office & Shortcuts

Item	Fact	□ Trick
System software	OS (Windows, Linux)	runs the machine
Application software	MS Word, Excel, browser	user tasks
MS Word file ext	.docx	word DOC
MS Excel file ext	.xlsx	X = eXcel
PowerPoint ext	.pptx	PPT
Excel: cell ref	column-letter + row-number (e.g. B5)	—
Excel: formula starts with	= sign	=SUM(A1:A5)

Shortcut	Action	Shortcut	Action
Ctrl+C / X / V	Copy / Cut / Paste	Ctrl+Z / Y	Undo / Redo
Ctrl+A	Select all	Ctrl+S	Save
Ctrl+B / I / U	Bold / Italic / Underline	Ctrl+F	Find
Ctrl+P	Print	Ctrl+Home	Go to start
F5	Refresh / Slideshow (PPT)	Alt+F4	Close window

3.3 Internet, E-mail & Cyber Security

Term	Meaning	□ Trick
HTTP / HTTPS	web protocol / **S = Secure**	padlock = HTTPS
URL	web address	http://...
ISP	Internet Service Provider	gives you internet
CC / BCC (e-mail)	Carbon Copy / **Blind** CC	BCC = hidden recipients
Virus / Worm / Trojan	malware types	Trojan = disguised as safe
Phishing	fake mail steals data	"fishing" for passwords
Firewall	blocks unauthorised access	network gatekeeper
Cookie	small file storing browse data	—

⚡ 60-Second Recall (cover the right column) — ASO scope

Prompt	Answer
km/h → m/s	× 5/18
Avg speed (equal distance)	$2S_1S_2/(S_1+S_2)$
CI − SI for 2 years	$P(R/100)^2$
$a+b+c=0 \Rightarrow a^3+b^3+c^3$	3abc
$x+1/x=k \Rightarrow x^3+1/x^3$	k^3-3k
Successive % a, b	a+b+ab/100
Clock angle	\
Sphere volume	$4/3 \cdot \pi r^3$
Cone slant l	$\sqrt{(r^2+h^2)}$
$\sin^2\theta+\cos^2\theta$	1
"only daughter of HIS mother"	his sister
Memory ladder	bit < Byte < KB < MB < GB < TB
HTTPS	secure web protocol
Save / Find shortcut	Ctrl+S / Ctrl+F

30H − 5.5M\

✅ **Study flow for ASO:** the whole exam is **Tier-I + Tier-II Paper-I** — so master **Part 1 (Quant)**, **Part 2 (Reasoning)**, and **Part 3 (Computer)** here, plus English & General Awareness. **Finance/Economics (Paper-III) and Statistics (Paper-II) are NOT in your syllabus.** Redraw the diagrams from memory — the boxed relationships are built to "snap" back in the exam.